



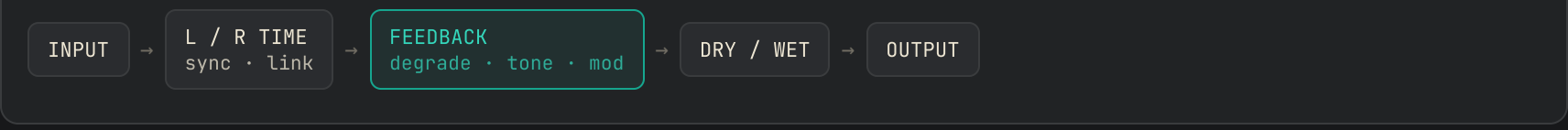
01 What It Is

Caverns is a stereo delay with independent left and right delay times, styled after analog bucket-brigade delay hardware. Degrade and the two modulation controls add the pitch wobble and lo-fi darkening bucket-brigades are known for, letting you dial in the perfect amount of character.

Low Cut and High Cut shape the tone of the feedback path itself, so each repeat darkens a little more than the last, the way a real bucket-brigade delay's repeats degrade over time.

02 Signal Path

L Time and R Time set how long each channel (Left and Right) waits before its first repeat. Everything else on the panel shapes what happens to the signal as it recirculates.



03 The Panel

TIMING CHARACTER MODULATION MIX

TIMING



Sync on/off · default off
Locks L Time and R Time to tempo-synced note divisions (L Division / R Division) instead of free-running milliseconds.



Link on/off · default off
Forces R Time to exactly follow L Time, so both channels always match.



L Time 1–2000 ms · default 350.0 ms
Left channel delay time. Unused while Sync is on.



R Time 1–2000 ms · default 350.0 ms
Right channel delay time. Unused while Sync or Link is on.

CHARACTER



Feedback 0–100% · default 53.8%
Delay regeneration amount, as in how many times it will repeat.



Degrade 0–100% · default 0.0%
Bucket-brigade-style lo-fi character and pitch wobble in the feedback path. At 0% the repeats stay clean. Turning it up adds the flutter and grit of aging analog hardware.



Low Cut 20 Hz–2 kHz · default 80 Hz
High-pass filter on the feedback path. Raise it to thin out the low end of each successive repeat.



High Cut 1–20 kHz · default 8000 Hz
Low-pass filter on the feedback path. Lower it to darken each repeat, the way real bucket-brigade hardware rolls off treble over successive passes.

MODULATION



Mod Speed 0–100% · default 45.5%
Rate of the pitch wobble that the Degrade control adds to the feedback path.



Mod Depth 0–100% · default 0.0%
Depth of the pitch wobble Degrade adds to the feedback path. At 0% the wobble has no effect regardless of Mod Speed.

MIX



Dry 0–100% · default 100%
Level of the untouched input.



Wet 0–200% · default 35%
Level of the delayed signal, independent of Dry. Can be pushed past unity for an exaggerated, obviously-effected sound.

A Bypass switch sits at the top right of the panel. It mutes the effect entirely, passing dry signal only, for quick A/B comparison.